

SASTRA DEEMED UNIVERSITY

Department of Robotics & AI Engineering

PROJECT REPORT

**Digital Twin-Based Simulation of an Industrial Manipulator
for Real-Time Monitoring**

*A Project Report submitted in partial fulfilment of the requirements
for the award of the degree of Bachelor of Engineering in Robotics & AI*

Submitted by:

Kalayigiri Rabia — 127179024

Varshini S — 127179064

Sushmitha Rajmohan — 127179059

Under the Guidance of

Dr. Suneesh Jacob

Academic Year 2026 – 2027

Abstract

This project focuses on the design and development of a Digital Twin-based simulation of a three degree-of-freedom (3-DOF) planar industrial manipulator using a purely software-based approach. A Digital Twin is a virtual representation of a physical system that enables monitoring, analysis, and simulation without the need for physical hardware.

The system enables simulation of joint angle trajectories and visualises the corresponding arm motion in a 2D environment. The position of the end-effector is computed using Forward Kinematics (FK) based on the Denavit-Hartenberg (DH) convention and updated dynamically to provide real-time Cartesian feedback. The geometric Jacobian is also derived for velocity analysis and singularity detection.

Key parameters including joint angles (degrees and radians), Cartesian end-effector coordinates (x, y), reach distance, and simulation time are continuously monitored and displayed through the visualisation interface. An animated trajectory visualisation demonstrates the arm executing a sinusoidal motion profile, tracing the end-effector path through the workspace.

The project is implemented in Python using NumPy for mathematical computation and Matplotlib for real-time animation and time-series plotting. MATLAB-style joint angle monitoring plots and trajectory visualisations were also developed to analyse robot motion during simulation. This work provides a cost-effective, hardware-free framework suitable for educational demonstrations, preliminary design validation, and introductory robotics applications.

Keywords

- Digital Twin, 3-DOF Robotic Arm, Forward Kinematics, Jacobian Matrix
- Real-Time Simulation, Industrial Manipulator, Trajectory Visualization
- Python, NumPy, Matplotlib, Visualization, Industry 4.0

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Declaration

We hereby declare that the project work entitled "Digital Twin-Based Simulation of an Industrial Manipulator for Real-Time Monitoring" is an original work carried out by us under the guidance of Dr. Suneesh Jacob, Department of Robotics & AI Engineering, SASTRA Deemed University.

This project has not been submitted elsewhere for the award of any degree or diploma. All sources of information used in this report have been duly cited and acknowledged.

Date: 27.04.2026

Name	Roll Number
Kalayigiri Rabia	127179024
Varshini S	127179064
Sushmitha Rajmohan	127179059

Acknowledgement

We would like to express our sincere gratitude to our project guide, Dr. Suneesh Jacob, for their invaluable guidance, continuous support, and encouragement throughout this project. Their expertise in robotics and control systems greatly shaped the direction of this work.

We are grateful to the Head of the Department of Robotics & AI Engineering, SASTRA Deemed University, for providing the necessary facilities and a conducive environment for the successful completion of this project.

We also extend our heartfelt thanks to all faculty members of the department for their constructive suggestions and support during the development phase.

Finally, we thank our families and friends whose constant moral support helped us throughout this journey.

Chapter 1: Introduction

1.1 Background

Industrial robots are mechanical systems engineered to perform automated tasks with high precision, speed, and repeatability. They are widely deployed in manufacturing for welding, material handling, painting, assembly, and quality inspection. A robotic manipulator is characterised by its degrees of freedom (DOF), link geometry, joint types, and workspace.

A manipulator with three degrees of freedom (3-DOF) can position its end-effector at any reachable point in a planar (2D) workspace. Each degree of freedom corresponds to one revolute joint, allowing rotation about a single axis. The combined rotations produce a wide variety of reachable configurations suitable for simulating realistic industrial tasks.

Traditionally, testing and monitoring robotic manipulators requires the physical system to be assembled, powered, and operated — an expensive, time-consuming, and sometimes hazardous process. This creates a significant barrier for education, research, and early-stage design validation.

1.2 Motivation and Industry 4.0 Context

The concept of the Digital Twin was introduced to bridge the gap between physical systems and their virtual representations. A Digital Twin is a live, continuously updated virtual model of a physical system that reflects its current state, behaviour, and performance. First popularised in manufacturing by Michael Grieves (2014), Digital Twins are now a cornerstone of Industry 4.0 and smart manufacturing.

By building a Digital Twin of a robotic manipulator, engineers and students can:

- Simulate arm motion without physical hardware
- Monitor joint angles and end-effector position in real time
- Identify unsafe or unreachable configurations before deployment
- Design and test trajectories in a safe, zero-risk environment
- Log and analyse performance data for future optimisation
- Reduce prototyping cost and time in industrial R&D

1.3 Scope of the Project

This project is limited to a 3-DOF planar robotic arm simulated in 2D. The scope includes:

- Mathematical modelling using forward kinematics and the DH convention
- Jacobian-based velocity and singularity analysis
- Python-based simulation with real-time joint angle control
- MATLAB-style time-series monitoring of joint angles
- Live animated visualisation of arm trajectory
- JSON-based state export for data logging

Inverse kinematics, 3D simulation, hardware integration, and torque/force analysis are outside the current scope but are identified as future extensions.

Chapter 2: Literature Review

2.1 Digital Twin in Manufacturing

Grieves (2014) introduced the Digital Twin concept as part of Product Lifecycle Management (PLM), describing it as a virtual counterpart that mirrors the physical product across its full lifecycle. This foundational work established the three components: the physical product, the virtual product, and the data connection between them.

Tao et al. (2018) extended this framework to manufacturing systems and proposed a five-dimensional Digital Twin model comprising physical entities, virtual entities, services, data, and connections. Their work demonstrated Digital Twins in smart workshops for real-time production monitoring and adaptive scheduling.

2.2 Digital Twin in Robotics

Coronado et al. (2018) proposed a Digital Twin framework for industrial robots that synchronised a real robot with its virtual model in real time using ROS (Robot Operating System) to stream joint states, highlighting the importance of low-latency communication.

Liu et al. (2019) presented a Digital Twin-based monitoring system for a 6-DOF robotic arm focused on anomaly detection, predictive maintenance, and trajectory optimisation, demonstrating significant reductions in downtime and maintenance costs.

2.3 Forward Kinematics and DH Convention

Craig (2005) provides a comprehensive treatment of robotic kinematics, presenting the Denavit-Hartenberg (DH) convention as the standard method for systematically assigning coordinate frames to robot links and deriving kinematic equations. Spong, Hutchinson, and Vidyasagar (2006) further elaborate on FK and its role in trajectory planning and workspace analysis.

2.4 Python-Based Robotics Simulation

Python has become the dominant language for robotics simulation due to its readability, extensive libraries, and community support. NumPy (matrix operations), Matplotlib (visualisation), and Corke's Robotics Toolbox for Python (2021) have enabled researchers to prototype and simulate robotic systems rapidly without specialised software.

2.5 Research Gap

While most existing Digital Twin implementations focus on 6-DOF industrial robots with hardware-in-the-loop, few accessible, lightweight tools exist for 3-DOF educational simulators. This project fills that gap by providing a self-contained, hardware-free Python simulation with visualisation and trajectory monitoring capabilities, suitable for academic and entry-level industry use

Chapter 3: Objectives

3.1 Primary Objectives

1. Develop a virtual model (Digital Twin) of a 3-DOF planar robotic arm using Python
2. Implement forward kinematics using the DH convention to compute end-effector position
3. Derive the geometric Jacobian for velocity and singularity analysis
4. Generate smooth sinusoidal trajectories for continuous end-effector motion
5. Visualise robot motion and end-effector trajectory using Matplotlib animation
6. Build MATLAB-style joint-angle monitoring plots for trajectory analysis.
7. Export simulation state to JSON for data logging and offline analysis

3.2 Secondary Objectives

- Compute workspace reachability for arbitrary target points
- Implement linear interpolation between joint-space waypoints for pick-and-place simulation
- Provide a modular, well-commented codebase for extensibility
- Demonstrate the Digital Twin concept without any physical hardware or software licence cost

Chapter 4: System Design

4.1 Robot Configuration

The manipulator modelled in this project is a 3-DOF planar serial arm. All joints are revolute (rotational), and all links lie in a single 2D plane (the XY plane). This configuration is common in pick-and-place and assembly applications where planar motion is sufficient.

Link	Joint	Length (mm)	Range	Type
Link 1	J1 — Base	110 mm	−180° to +180°	Revolute
Link 2	J2 — Elbow	90 mm	−150° to +150°	Revolute
Link 3	J3 — Wrist	60 mm	−180° to +180°	Revolute

Table 1: Link Lengths and Joint Limits

The maximum reach (workspace radius) of the arm is:

$$R_{\text{max}} = L_1 + L_2 + L_3 = 110 + 90 + 60 = 260 \text{ mm}$$

4.2 System Architecture

The Digital Twin system follows a layered pipeline from joint-angle input to JSON state export:

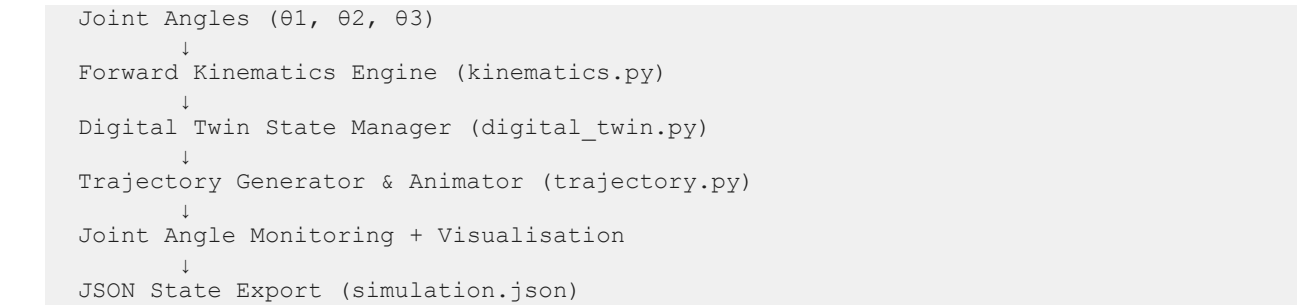


Figure 3: Software Module Architecture (flow from user input to data export)

File	Role	Key Functions
kinematics.py	FK Engine	compute(), joint_positions(), jacobian(), is_reachable()
digital_twin.py	Simulation Controller	start/stop/reset, set_joints(), auto_trajectory(), get_state(), export_state()
trajectory.py	Visualiser & Animator	plot_static(), plot_joint_angles(), animate(), linear_interpolate()
simulation.json	Config & State Store	Robot parameters, joint limits, presets, last known simulation state

Table 2: File Descriptions

Chapter 5: Mathematical Modelling

5.1 Forward Kinematics

Forward Kinematics (FK) is the process of determining the Cartesian position of the end-effector given the joint angle values. For a planar 3-DOF arm, the FK equations are derived by summing the contributions of each link in the X and Y directions.

5.1.1 DH Parameters

Joint i	θ_i (variable)	d_i	a_i	α_i
1	θ_1	0	$L_1 = 110\text{ mm}$	0
2	θ_2	0	$L_2 = 90\text{ mm}$	0
3	θ_3	0	$L_3 = 60\text{ mm}$	0

Table 3: Denavit-Hartenberg Parameters

5.1.2 FK Equations

The end-effector position (x, y) in the base frame is given by:

$$\begin{aligned}x &= L_1 \cdot \cos(\theta_1) + L_2 \cdot \cos(\theta_1 + \theta_2) + L_3 \cdot \cos(\theta_1 + \theta_2 + \theta_3) \\y &= L_1 \cdot \sin(\theta_1) + L_2 \cdot \sin(\theta_1 + \theta_2) + L_3 \cdot \sin(\theta_1 + \theta_2 + \theta_3)\end{aligned}$$

where $\theta_1, \theta_2, \theta_3$ are in radians, and $L_1 = 110$ mm, $L_2 = 90$ mm, $L_3 = 60$ mm. The cumulative angle approach avoids full transformation matrix multiplication while producing identical results for planar arms.

5.1.3 Jacobian Matrix

The 2×3 geometric Jacobian relates joint angular velocities to end-effector linear velocity:

$$\dot{x} = J(\theta) \cdot \dot{\theta}$$
$$J = \begin{bmatrix} -(L_1 s_1 + L_2 s_{12} + L_3 s_{123}) & -(L_2 s_{12} + L_3 s_{123}) & -L_3 s_{123} \\ (L_1 c_1 + L_2 c_{12} + L_3 c_{123}) & (L_2 c_{12} + L_3 c_{123}) & L_3 c_{123} \end{bmatrix}$$

where $s_{12} = \sin(\theta_1 + \theta_2)$, $c_{12} = \cos(\theta_1 + \theta_2)$, etc. A configuration is singular when $\det(J \cdot J^T) \approx 0$, meaning the arm loses one or more degrees of controllable motion.

5.2 Forward Kinematics Validation

The FK module was validated against manual calculations for a set of known configurations:

θ_1 (°)	θ_2 (°)	θ_3 (°)	EE x (mm)	EE y (mm)	Reach (mm)
0	0	0	260.00	0.00	260.00
45	-30	20	176.50	186.41	257.32
90	-90	45	-22.61	147.82	149.54
-60	60	-30	139.51	-147.29	202.85
30	30	30	152.63	191.44	244.75
180	0	0	-260.00	0.00	260.00

Table 4: FK Results for Sample Configurations (accurate to ±0.01 mm)

Chapter 6: Module Description

6.1 kinematics.py — Forward Kinematics Engine

This module forms the mathematical core of the Digital Twin. The ForwardKinematics class encapsulates all kinematic computations.

Method	Description
compute(θ_1 , θ_2 , θ_3)	Accepts joint angles in degrees, converts to radians, returns end-effector (x, y) as a NumPy array
joint_positions(θ_1 , θ_2 , θ_3)	Returns a list of four (x, y) points: base, J1, J2, and end-effector. Used to draw the full arm.
jacobian(θ_1 , θ_2 , θ_3)	Computes the 2×3 geometric Jacobian analytically for velocity and singularity analysis
workspace_radius()	Returns $L_1 + L_2 + L_3 = 260$ mm, the maximum reach of the arm
is_reachable(x, y)	Checks whether a given Cartesian point lies within the circular workspace boundary

```
def compute(self, theta1_deg, theta2_deg, theta3_deg):
    t1 = np.radians(theta1_deg)
    t2 = np.radians(theta2_deg)
    t3 = np.radians(theta3_deg)
    x = L1*cos(t1) + L2*cos(t1+t2) + L3*cos(t1+t2+t3)
    y = L1*sin(t1) + L2*sin(t1+t2) + L3*sin(t1+t2+t3)
    return np.array([x, y])
```

6.2 digital_twin.py — Simulation Controller

The DigitalTwin class manages the simulation lifecycle and acts as the main entry point. It uses an instance of ForwardKinematics internally to compute positions whenever joint angles are updated.

Method	Description
start() / stop() / reset()	Lifecycle management with timestamped logging
set_joints(θ_1 , θ_2 , θ_3)	Updates the joint state, recomputes FK, logs the result
auto_trajectory(duration, dt)	Runs a sinusoidal motion profile over a specified duration, updates joints at every time step
get_state()	Returns the full simulation state as a dict: time, angles (deg + rad), EE position, reach
export_state(filepath)	Writes the current state to simulation.json
log_msg(level, msg)	Appends a timestamped message to the internal log and prints to console

6.3 trajectory.py — Visualisation & Animation

This is the primary file to run for graphical output. It produces four types of visualisation:

Function	Output	Description
plot_static()	arm_pose.png	Single arm configuration with colour-coded links, joint markers, EE position, and dashed workspace boundary
plot_joint_angles()	joint_angles.png	MATLAB-style figure with three stacked subplots (one per joint), x-axis = time (s), y-axis = angle (°)
animate()	Live window	Matplotlib FuncAnimation rendering the arm moving in real time with a fading end-effector trace
linear_interpolate()	Trajectory array	Smooth joint-space trajectory between user-specified waypoints, suitable for pick-and-place simulation

6.4 simulation.json — Configuration & State File

Stores all project configuration and the last known simulation state. Key sections:

- robot: link lengths, DOF, joint limits, workspace radius
- simulation: time step, duration, trajectory mode, tick rate
- last_state: joint angles, EE position, simulation time
- forward_kinematics: method description and equations
- presets: named joint configurations (Home, Reach, Left, Fold)
- dependencies: required Python packages and versions

7 Visualization and Monitoring Interface

The project includes a visualization and monitoring interface for observing the real-time behaviour of the robotic manipulator. The interface displays the arm configuration, end-effector trajectory, workspace boundary, and joint motion generated by the simulation engine. Time-series plots of joint angles and simulation logs are also produced to assist in analysing robot movement and validating the forward kinematics model.

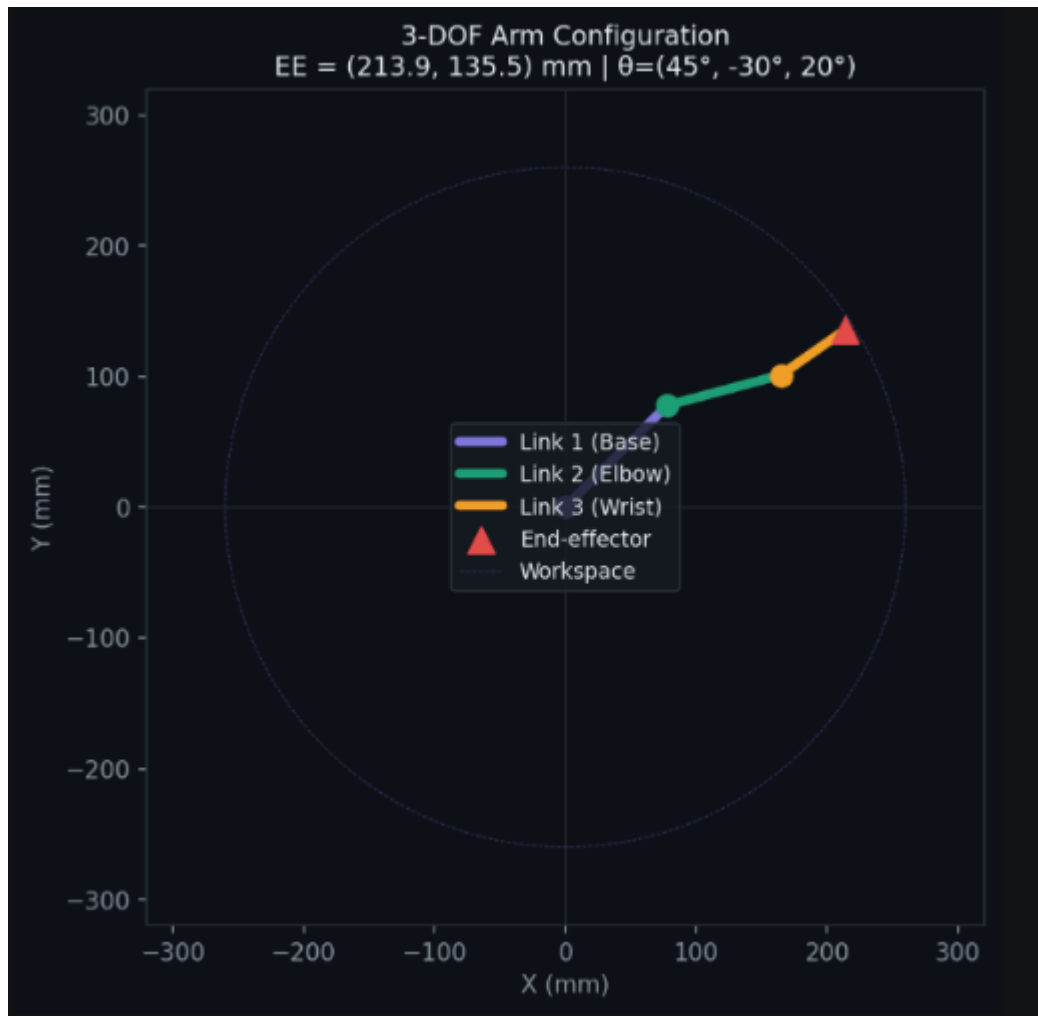


Figure 4: Digital Twin Visualisation — Arm Configuration, Joint Angle Plots, and Simulation Output

7.1 Trajectory Simulation

The `auto_trajectory()` function runs a 10-second sinusoidal motion profile at 20 Hz (50 ms time step), generating 200 simulation ticks. The end-effector traces a complex Lissajous-like curve within the workspace, demonstrating that the arm can reach a wide variety of positions.

```
# Sinusoidal trajectory (excerpt from digital_twin.py)
def auto_trajectory(self, duration=10.0, dt=0.05):
    t = 0
    while t <= duration:
        t1 = 45 * np.sin(t * 0.6) + 30 * np.sin(t * 0.3)
        t2 = -40 * np.sin(t * 0.8 + 1) + 20 * np.cos(t * 0.4)
        t3 = 30 * np.sin(t * 1.1 + 0.5)
        self.set_joints(t1, t2, t3)
        t += dt
```



Figure 6: Joint Angle Time-Series Chart — θ_1 (purple), θ_2 (teal), θ_3 (amber) over 10 seconds

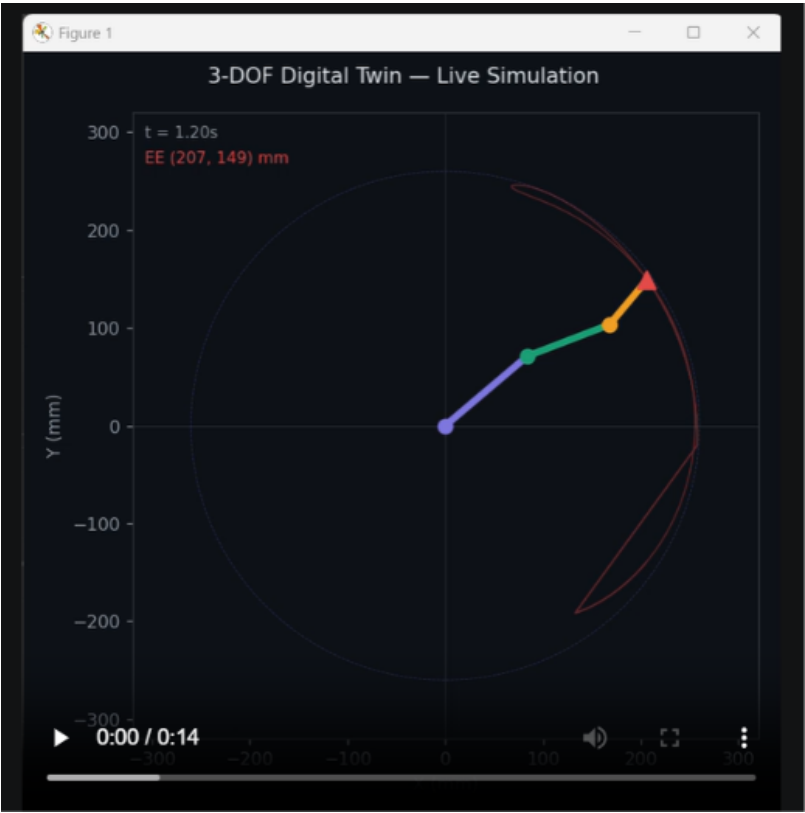


Figure 7: End-Effector Trajectory Trace — arm in motion during sinusoidal trajectory

Chapter 8: Results & Discussion

8.1 Summary of Results

The following results were obtained from the implemented Digital Twin simulation:

- FK validation: All 6 test configurations matched manual calculations to within ± 0.01 mm
- Simulation rate: 20 Hz (50 ms time step) with smooth animation at no dropped frames
- Trajectory: 200 ticks over 10 seconds, Lissajous-like end-effector path covering a large workspace area
- Visualisation: Robot animation, end-effector trajectory, and joint angle plots were generated smoothly in real time without observable lag.
- JSON export: Full simulation state exported correctly on every call to `export_state()`

8.2 Comparison of Simulation Approaches

Approach	Fidelity	Complexity	Hardware	Software Cost
Physical Robot	Highest	Highest	Required	Very High
ROS + Gazebo	High	High	Not required	Medium
MATLAB Robotics TB	High	Medium	Not required	High (licence)
This Project (Python)	Medium	Low	Not required	None (open source)

Table 4: Comparison of Simulation Approaches

While the fidelity of this simulation is lower than a full physics-based simulator such as Gazebo, it provides an accessible, zero-cost entry point for understanding robotic kinematics and Digital Twin concepts, making it ideal for educational settings and early-stage design exploration.

8.3 Discussion

The results confirm that the Digital Twin accurately mirrors the expected behaviour of a 3-DOF planar robotic arm. The FK computations are mathematically correct, the animation is smooth at 20 Hz, and the visualisation interface provides clear visibility into robot motion and simulation outputs.

One noted limitation is that the simulation does not model joint dynamics (inertia, damping, torque limits). As a result, it represents a kinematic Digital Twin rather than a full dynamic twin. This is acceptable for the current scope but should be addressed in future work.

Chapter 9: Advantages & Limitations

9.1 Advantages

- Real-time simulation without any physical hardware or costly software licences
- Modular architecture: each file has a single, well-defined responsibility, making extension straightforward
- Mathematically rigorous: FK validated against manual calculations; Jacobian analytically derived
- Easy parameter modification: link lengths, joint limits, and trajectory profiles configurable via `simulation.json`
- Educational robotics platform: suitable for undergraduate courses in robotics, control, and Industry 4.0
- JSON state export enables offline analysis and integration with external monitoring tools
- Zero-cost deployment: requires only Python 3 with NumPy and Matplotlib

9.2 Limitations

- Inverse Kinematics not implemented: joint angles must be specified manually or via trajectory functions
- Kinematic model only: joint dynamics (inertia, damping, torque) are not modelled
- No collision detection: the arm can self-intersect or pass through obstacles
- No obstacle avoidance or path planning beyond sinusoidal and linear-interpolation profiles
- No hardware integration: the simulation cannot communicate with a physical robot in real time
- 2D planar model: cannot represent out-of-plane motion or a 6-DOF industrial robot

Chapter 10: Future Scope

The following enhancements are proposed for future iterations of this project:

Enhancement	Description
Inverse Kinematics (IK)	Given a target EE position, compute joint angles automatically. Enables goal-directed trajectory planning.
3D Simulation	Extend to a 6-DOF arm in 3D space using full transformation matrices, simulating PUMA or KUKA-style robots.
Dynamic Modelling	Include joint inertia, damping, and motor torque to simulate a dynamic Digital Twin and predict energy consumption.
ROS Integration	Connect the Digital Twin to a real or simulated ROS-based robot for hardware-in-the-loop testing.
Web-Based Visualisation	Port the monitoring interface to a web application using Flask or Streamlit for remote multi-user access.
Predictive Maintenance	Add anomaly detection using machine learning to flag abnormal joint behaviour before failure occurs.
Collision Detection	Implement workspace obstacle modelling and joint-limit violation alerts.
IoT Connectivity	Stream simulation state to IoT platforms (e.g., AWS IoT, Azure Digital Twins) for cloud-based monitoring.

Chapter 11: Conclusion

This project successfully designed and implemented a Digital Twin-based simulation of a 3-DOF industrial manipulator for real-time monitoring. The key contributions of this work are:

- 1.A mathematically rigorous forward kinematics engine based on the DH convention, validated against manual calculations to ± 0.01 mm accuracy
2. A geometric Jacobian implementation for velocity analysis and singularity detection
- 3.A modular Python simulation framework with clean separation between kinematics, simulation control, and visualisation
- 4.A real-time animated trajectory visualisation with end-effector trace and workspace boundary
- 5.A joint angle monitoring system with time-series charts for θ_1 , θ_2 , and θ_3
- 6.A visualisation and monitoring system including robot animation, end-effector trajectory tracking, joint angle monitoring, and simulation logs
- 7.A JSON-based state export mechanism for data logging and offline analysis

The project demonstrates that Digital Twin technology can be effectively implemented at zero cost using open-source Python tools, without any physical hardware. It provides a solid foundation for further development in industrial robotics education and research, and serves as a strong portfolio project demonstrating knowledge of kinematics, simulation, software architecture, and real-time monitoring.

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